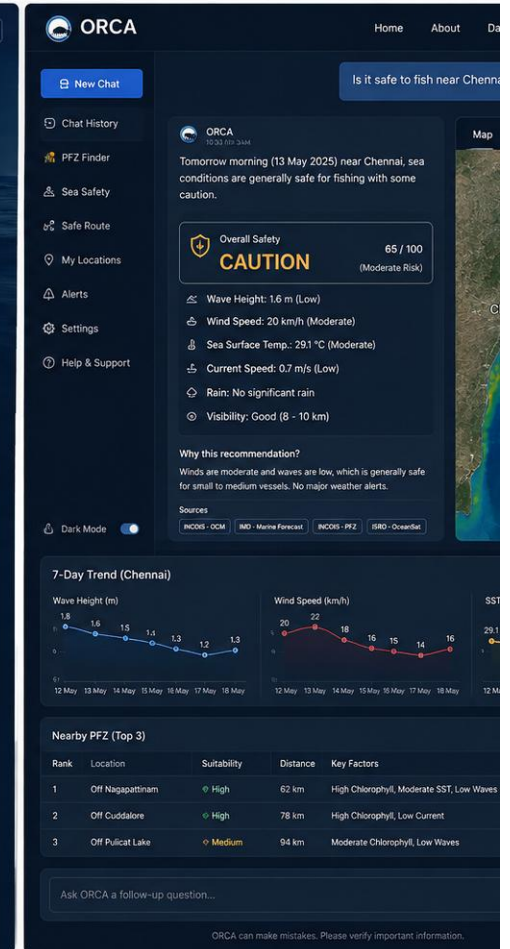
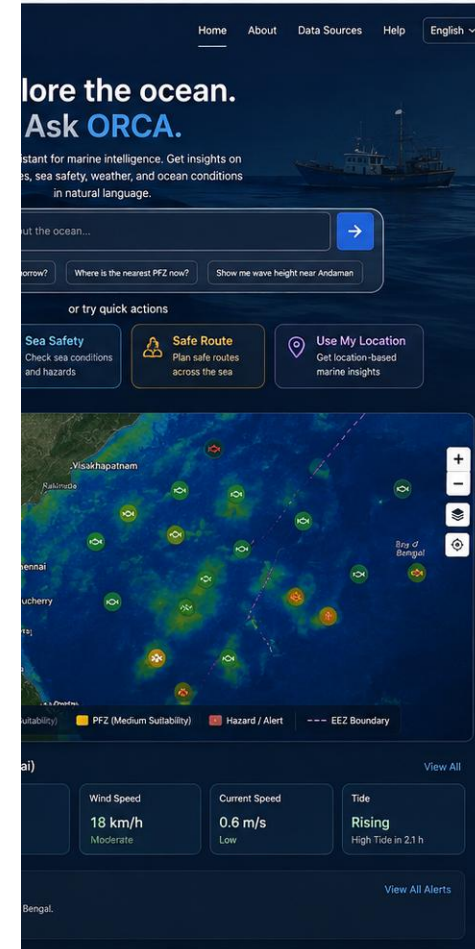


SMART INDIA HACKATHON 2026

TITLE PAGE

- Problem Statement ID – **SIH26176**
- Problem Statement Title – ORCA Marine Ecosystem Reasoning with Collaborative Agents.
- Theme – Miscellaneous
- PS Category – Software
- Team ID – [Team ID]
- Team Name – **CodeCatalysts**



ORCA — Agentic AI for Ocean Intelligence

DETAILED SOLUTION

- **Conversational Marine Decision Assistant:** Users query ocean conditions in natural language via Text or Vernacular Voice.
- **8 Specialized AI Agents:** Collaborative multi-agent team coordinates planning, analytics, weather, safety, routing, and synthesis.
- **Multi-Source Grounded Telemetry:** Integrates live satellite (NOAA/ISRO), weather models (Open-Meteo), and in-situ buoys (INCOIS).
- **Explainable & Evidence-Based:** Strict Zero-Hallucination guarantee with full data provenance citations & sensor latency ledgers.
- **Complete Maritime Cockpit HUD:** Unified chat, native 60 FPS Windy canvas fluid simulation, turn-by-turn steerage, and GMDSS SOS beacon.

HOW IT ADDRESSES THE PROBLEM

- **Overcomes Fragmented Data:** Unifies scattered data from IMD, INCOIS, ISRO, NOAA, Coast Guard into one clear interface.
- **Eliminates Manual Search:** Replaces multi-portal search with instant, authoritative natural-language answers in < 1 second.
- **Contextual & Location-Aware:** Automatically detects craft type (artisanal skiffs vs commercial trawlers) and geocodes 35+ coastal ports.
- **Bridges Technical Complexity:** Converts raw GRIB rasters and numerical buoy telemetry into simple, life-saving advice.
- **Delivers Multi-Modal Outputs:** Outputs interactive maps, Douglas sea-state scales, Beaufort wind forces, steerage, and evidence tables.

INNOVATION & UNIQUENESS

- **8 Collaborative AI Agents:** 1 Planning + 7 Specialized Domain Agents working with asynchronous task decomposition.
- **Native Indian Language Voice:** Full Speech-to-Text & TTS in 12+ coastal languages (Tamil, Malayalam, Hindi, Telugu, Marathi, Bengali).
- **60 FPS Fluid Canvas Simulator:** Windy-style real-time particle streams, wave swell wavefronts, and SST heatmaps over Leaflet maps.
- **Deterministic Safety Engines:** Pure mathematical scoring (0–100) and Cape Comorin fairway routing with 0% land crossing guarantee.
- **Emergency GMDSS SOS Beacon:** Automated Coast Guard MRCC routing (Kochi, Mumbai, Chennai), AIS ship proximity, and PDF receipts.

FACTS & FIGURES — WHY IT MATTERS

11,099 km

India's revised coastline
(Survey of India 2025)

18%

Population in 72 coastal
districts (Sagarmala)

~80 Million

People depend on the sea
for livelihoods (Govt India)

28.06 Million+

Fishermen in India
(Min. of Fisheries 2024)

95% (vol) / 70% (val)

India's trade moved via
maritime routes (MoPSW)

₹62,408 Crore

Seafood export earnings
(MPEDA FY 2024-25)

WHO NEEDS MARINE INTELLIGENCE?

- **Fishing Communities:** PFZ discovery, 20–30% fuel savings, rough sea safety clearance.
- **Researchers & Academia:** Direct access to SST, chlorophyll-a thermal fronts & thermoclines.
- **Government & Authorities:** Real-time hazard alerts, IMD port signals 1–11 & Coast Guard SAR.
- **Maritime Operators:** Nautical routing avoiding shallow shoals & deep-water MPA corridors.
- **Environmental Stakeholders:** Geofence protection for Marine Protected Areas (Gahirmatha, Gulf of Mannar).

TECHNICAL APPROACH

ORCA
n

IMPLEMENTATION FLOW

1

USER INPUT

Question in natural language (Text/Voice) + optional GPS location or port.

2

PLANNING

Planning Agent classifies intent, geocodes 35+ coastal ports, extracts timeframe.

3

8 AI AGENTS

Orchestrator dynamically activates required domain-specific specialists.

4

DATA RETRIEVAL

Parallel multi-feed ingestion: Open-Meteo, NOAA GHRSST, ISRO OCM-3, Buoys.

5

VALIDATE & ANALYSE

Data normalization, sensor quality check, <12h freshness validation.

6

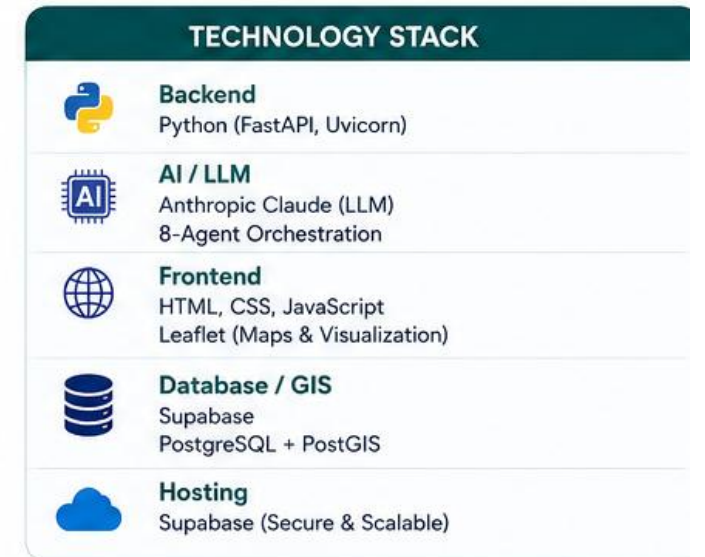
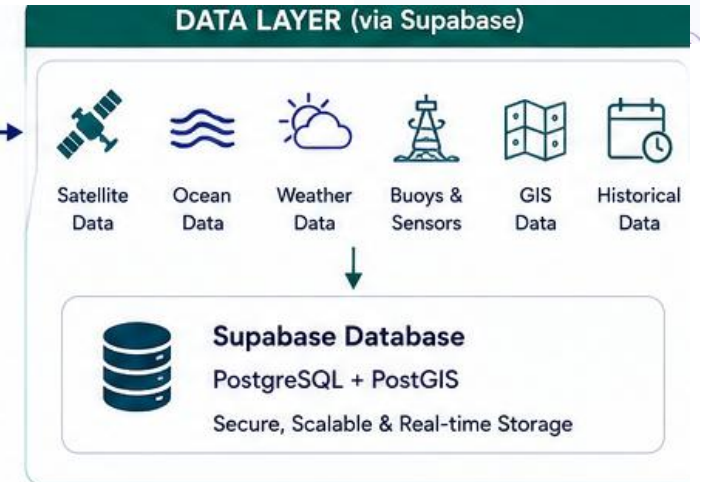
REASON & ASSESS

Deterministic Risk Engine (0–100), Fairway Route Engine, and PFZ Engine.

7

OUTPUT

Synthesized explanation + evidence citations + 60 FPS Windy map + PDF receipts.



FEASIBILITY AND VIABILITY

FEASIBILITY

- **Reliable Data Sources** — Open-Meteo, NOAA CoastWatch GHRSSST L4, ISRO MOSDAC OCM-3, INCOIS Buoys, IMD.
- **Proven Technologies** — Python 3.13, FastAPI, PostGIS, Leaflet.js, React 19, PWA offline caching.
- **Modular Architecture** — 8 independent domain agents + plug-and-play connector interfaces.
- **Scalable Infrastructure** — Supabase cloud backend + embedded zero-dependency spatial index.
- **Feasible & Tested MVP** — 100% automated test suite passing (34/34 tests across SIH scenarios).

VIABILITY

- **Multi-User Value** — Supports artisanal canoes, mechanized trawlers, coast guards, port masters.
- **Continuous Updates** — Dynamic satellite passes (GHRSSST L4 daily, Open-Meteo hourly).
- **Expandable Services** — Extensible to carbon credits, mariculture, and deep-sea mining.
- **Scalable Deployment** — Pilot tested in Kochi, Mumbai, Chennai; scalable to all 72 coastal districts.

CHALLENGES & RISKS

HOW WE HANDLE THEM (ORCA SOLUTION)

1. Different data formats and sources

→ Unified Data Normalization Engine (Standard Pydantic v2 schemas)

2. Data freshness and time lags

→ Timestamps + freshness engine (+8 risk penalty if telemetry > 12h)

3. Conflicting or incomplete data

→ Cross-source validation (In-situ OMNI buoys cross-check satellite grid)

4. LLM hallucination and incorrect reasoning

→ Strict Grounded Heuristics (LLM only for intent; math strictly deterministic)

5. Large-volume spatial queries at sea

→ Embedded Ray-Casting & Haversine spatial index (< 5ms zero-latency queries)

6. Accuracy of risk & route recommendations

→ IMO/WMO rules (Douglas/Beaufort scales) & Cape Comorin fairway corridors

7. External API downtime / zero signal at sea

→ PWA Offline Service Worker + Pre-seeded local fallback cache

8. User privacy & emergency distress security

→ Stateless session tokens & encrypted GMDSS SHA-256 dispatch tracking

IMPACT AND BENEFITS

WHO IS IMPACTED?



Find better fishing zones
Plan safer trips
Receive timely alerts
Improve daily decisions

FISHING COMMUNITIES

- Get advisories on potential fishing zones (PFZ).
- Plan safer, 20–30% fuel-optimized voyages.
- Drastic drop in capsizing & rough sea accidents.



Access integrated marine data
Analyze trends & patterns
Support scientific studies
Faster insights for research

RESEARCHERS & ACADEMIA

- Instant access to multi-source satellite oceanography.
- Accelerates research on upwelling & thermoclines.
- Real-time data for marine climate modeling.



Monitor marine conditions
Issue timely advisories
Manage hazards & disasters
Stronger decision support

• WIDER BENEFITS •

GOVERNMENT & AUTHORITIES

- Real-time alerts for cyclones, squalls & high waves.
- Automated IMD port signals 1–11 dissemination.
- Coast Guard SAR emergency beacon integration.



Plan optimal & safe routes
Check weather & sea conditions
Reduce delays & operational cost
Improve voyage efficiency

MARITIME OPERATORS

- Plan routes with weather & sea-state insights.
- Avoid restricted zones (MPA / IMBL borders).
- Reduce operational expenditure & transit delays.



Monitor sensitive marine areas
Support conservation efforts
Assess environmental changes
Promote sustainable activities

ENVIRONMENTAL BODIES

- Monitor sensitive marine sanctuaries (Gahirmatha).
- Protect Olive Ridley nesting & coral reefs.
- Promote sustainable fishing & Blue Economy.

WIDER BENEFITS

SOCIAL

- Inclusivity via 12+ native Indian languages.
- Voice-first interface for low-literacy fishermen.
- Drastically improves coastal safety awareness.

ECONOMIC

- 20–30% fuel savings per fishing voyage.
- Minimizes catch spoilage with targeted PFZ runs.
- Lowers avoidable maritime search & rescue expenditure.

ENVIRONMENTAL

- Reduces maritime carbon footprint.
- Prevents illegal incursions into Marine Sanctuaries.
- Promotes sustainable, balanced fishing practices.

SCALABILITY

- **REGIONAL → NATIONAL:** Start with West Coast (Kochi/Mumbai) pilot; scale to all 7,516+ km coastline & island territories.
- **MODULAR ARCHITECTURE:** Add new satellite channels (NISAR, Oceansat-4) & sensors without core system rebuild.
- **CLOUD-READY STACK:** Dockerized microservices & Supabase cloud ready to scale across millions of daily coastal queries.
- **MORE DATA → MORE VALUE:** Fishermen catch logs and in-situ buoy networks create self-improving predictive intelligence.

RESEARCH AND REFERENCES

RESEARCH BASIS

• Marine Earth Observation:

Multi-spectral satellite radiometry, SST thermal fronts ($\Delta T/\text{km}$), and chlorophyll-a upwelling dynamics.

• Operational Oceanography:

Douglas Sea State Scale (0–9), Beaufort Wind Force Scale (0–12), Ekman Transport, and thermocline stratification.

• Geospatial & Marine Navigation:

Haversine geodesic math, Ray-Casting point-in-polygon containment, and Traffic Separation Schemes (TSS).

• Multi-Agent Collaborative AI:

Hierarchical intent decomposition, deterministic agent delegation, and zero-hallucination execution.

• Conversational Marine UX:

Voice-first interaction in vernacular Indic scripts, high-contrast marine telemetry HUDs, and offline caching.

• Evidence & Provenance Modeling:

End-to-end audit ledgers, sensor latency tracking, and scientific citation standards.

DATA SOURCE GROUPS

• Satellite Earth Observation

ISRO Oceansat-3 OCM, NOAA CoastWatch GHRSSST L4 OSTIA, Copernicus Sentinel-3 OLCI.

• Marine Physical Parameters

Sea Surface Temperature (SST), Chlorophyll-a concentration, Wave height, Swell period, Ocean surface currents.

• Meteorological Intelligence

Wind vectors & gusts, Barometric pressure, GFS & ECMWF marine forecast models.

• Safety & Hazard Bulletins

INCOIS Ocean State Forecast, IMD Marine Weather, IMD Damini Lightning, IMD Port Warning Signals (1–11).

• Maritime GIS Boundaries

National Hydrographic Office, Indian EEZ (200 NM), IMBL (Sri Lanka/Pakistan), Marine Protected Areas (MPAs).

• Historical Marine Baselines

INCOIS historical PFZ calibrations, CMFRI marine fish landings (2015–2024), GEBCO ocean soundings.

REFERENCE AREAS & OFFICIAL BODIES

• **INCOIS:** Indian National Centre for Ocean Information Services (MoES, Hyderabad) — PFZ Advisories & Ocean State Forecasts.

• **ISRO MOSDAC:** Meteorological & Oceanographic Satellite Data Archival Centre (SAC, Ahmedabad) — Oceansat & INSAT-3D.

• **IMD:** India Meteorological Department (MoES, New Delhi) — Cyclone Warnings & Port Danger Signals 1–11.

• **Indian Coast Guard:** Maritime Rescue Coordination Centres (MRCC Kochi, Mumbai, Chennai, Port Blair) — SAR Protocols.

• **NOAA CoastWatch:** National Oceanic and Atmospheric Administration — Global GHRSSST Level-4 Sea Surface Temperature.

• **Survey of India & NIOT:** National Institute of Ocean Technology — OMNI Deep-Sea Buoys & Coastal Tide Gauges.

• **CMFRI & MPEDA:** Central Marine Fisheries Research Institute & Marine Products Export Development Authority — Catch & Export Data.

Final submission documents list exact API endpoints, dataset DOIs, and official government research citations used in implementation.